AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM358
Machine Learning & Artificial intelligence

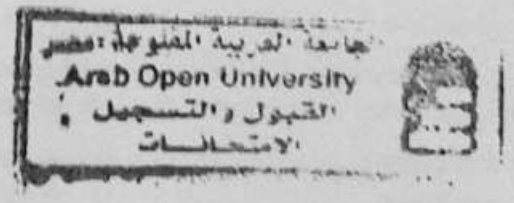
Midterm Examination (MTA)
Summer Semester 2025/2026

Time Allowed: 120 minutes

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Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.



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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 ..... are the means by which robots move and change the shape of their bodies.

A Sonar sensors
B GPS
C Effectors
D proprioceptive sensors

2 .....can be defined as "is a tiny 'world' comprising an (infinitely large) flat table on which sit a set of children's building blocks. "

A Block Chain
B Block world
C smart world
D None of the above

3 The technical term for when completing one sub-plan undoes the achievements of another is .....

A Block world
B planning
C clobbering
D None of the above

4 Backgammon is an example of ..... games

A both of the above
B stochastic
C deterministic
D None of the above

5 ..... can apply their effectors further afield than anchored manipulators can, but their task is made harder because they don't have the rigidity that the anchor provides.

A Manipulators
B Mobile robot
C mobile manipulator
D None of the above

6 ..... can be defined as "Given a set of goals, construct a sequence of actions that achieves those goals "

A Planning
B Performing
C All of the above
D Acting

7 .....can be defined as 'the appearance of novel and/or unpredictable, regular patterns or laws arising in systems from interactions between their simple, low-level components'

A selection
B emergence
C interaction
D adaptation

8 ..... knowledge deals with factoid questions

A Procedural
B Declarative
C Instructional
D All of the above

9 ..... is the set of states that we can be in.

A state space
B action set
C goal state
D None of the above

10 ..... is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators

A A machine
B A computer
C An Agent
D None of the above

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

A. Discuss the factors that contribute to the number of visited states in a depth-first search algorithm. [5 marks]

B. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	A	B	B	B	C	C	C	D	D	D	E	F	F	G	G	H	H	I	I	J	M	M
t	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Given the aforementioned tree, apply BFS to traverse the tree showing all agenda content and traced paths until reaching the goal state if exist assuming that the goal state is L.

Question 3: (15 marks)

A. Discuss the advantages and limitations of using simplified worlds like Blocks World for planning tasks in AI. [5 marks]

B. Having the following XO sequence of states a long with their values

- O O - - - X X -	0.500
- O O - - X X X -	0.533
- O O - - X X X O	0.543
X O O - - X X X O	0.501
X O O - O X X X O	0.540
X O O X O X X X O	1.000

Assume a learning rate of 0.86 what will be updated values adopting gradient-based state value update with each move.[10 marks]

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Question 4: (15 marks)

A. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assume s, t in order, for a given tree as:

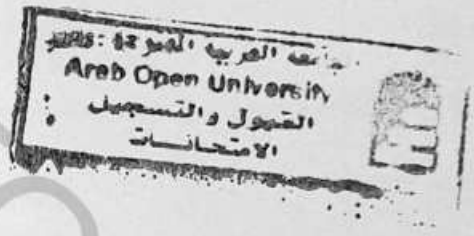
s	A	A	B	B	C	C	C	C	E	G	G	G	H	H	H
t	B	C	2	E	13	G	H	8	45	17	5	35	20	41	6

Given that tree, use α-β pruning minimax algorithm to evaluate each node in the tree.

B. Explain the role of operators in a planning system.

End of Questions

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